

Arithmetic operations on functions

Adding, subtracting, multiplying and dividing functions — and the one rule about the domain that decides every answer.

LESSON 13–14

2 × 40 MIN

ALGEBRA 10, §1.4

P1 · 2.3

LESSON CLOCK

6'

12'

10'

14'

8'

5'

Four new functions from two old ones	6 min
The domain rule	12 min
Division — the extra condition	10 min
Worked practice	14 min
Consolidation	8 min
Homework	5 min

LEARNING OBJECTIVES

- Form $f + g$, $f - g$, $f \cdot g$ and f/g from two given functions.
- Find the domain of a combined function as the intersection of the domains.
- Exclude the zeros of the denominator when dividing.
- Evaluate a combined function at a point.

TEXTBOOK

National	pp. 42–47
Cambridge	Pure Mathematics 1, pp. 37–41
Workbook	P1 Exercise 2C

01

Explanation

Four new functions from two

Given f and g , four new functions are defined pointwise — apply both rules to the same input, then combine the two outputs:

$$(f + g)(x) = f(x) + g(x) \cdot (f - g)(x) = f(x) - g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x) \cdot \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Nothing new is happening to the arithmetic. Everything new is happening to the domain.

The domain rule

BOTH RULES MUST ACCEPT THE INPUT

An input is allowed in $f + g$ only if it is allowed in f **and** in g . So the domain of the combination is the **intersection** of the two domains.

With $f(x) = \sqrt{x}$ (domain $x \geq 0$) and $g(x) = \frac{1}{x - 4}$ (domain $x \neq 4$), the sum has domain $x \geq 0$ and $x \neq 4$.

SIMPLIFYING DOES NOT WIDEN A DOMAIN
 $f(x) = \sqrt{x}$ and $g(x) = -\sqrt{x}$ give $(f + g)(x) = 0$ — but only for $x \geq 0$. The zero function on $\mathbb{R}$ is a different function.

Division needs one more exclusion

For $\frac{f}{g}$ the input must be in both domains **and** must not make g zero. Finding those zeros is usually the real work of the question.

domain of $\frac{f}{g} = (\text{dom } f) \cap (\text{dom } g) \text{ minus } \{x : g(x) = 0\}$

$F(X)$	$G(X)$	$(F/G)(X)$	DOMAIN
$x + 1$	$x - 2$	$\frac{x + 1}{x - 2}$	$x \neq 2$
x^2	$x^2 - 1$	$\frac{x^2}{x^2 - 1}$	$x \neq \pm 1$
$\sqrt{x}$	$x - 3$	$\frac{\sqrt{x}}{x - 3}$	$x \geq 0, x \neq 3$

02 Worked examples

EXAMPLE 1 $f(x) = x + 2$, $g(x) = x^2 - 4$. Find $(f + g)(3)$ and $(f \cdot g)(x)$.

① $f(3) = 5$, $g(3) = 5$

② $(f + g)(3) = 10$

③ $(f \cdot g)(x) = (x + 2)(x^2 - 4)$

④ $= (x + 2)^2(x - 2) = x^3 + 2x^2 - 4x - 8$

ANSWER

10 and $x^3 + 2x^2 - 4x - 8$

EXAMPLE 2 $f(x) = \sqrt{x - 1}$, $g(x) = x - 5$. Find the domain of $\frac{f}{g}$.

① Domain of f : $x \geq 1$

② Domain of g : $\mathbb{R}$

③ Intersection: $x \geq 1$

④ Remove the zero of g : $x = 5$

ANSWER

$x \geq 1$, $x \neq 5$

EXAMPLE 3 $f(x) = 3x$, $g(x) = x - 1$. Solve $(f - g)(x) = 9$.

① $(f - g)(x) = 3x - (x - 1)$
Bracket the whole of g .

② $= 2x + 1$

③ $2x + 1 = 9 \Rightarrow x = 4$

ANSWER

$$x = 4$$

03 Interactive model

Pick two functions and see the four combinations plotted together.

Combining functions

$f(x) = 2x$, $g(x) = x + 5$. Then $(f + g)(1)$ is:

$(f - g)(x)$ for the same pair is:

Domain of $f + g$ is the ____ of the domains:

$f(x) = 1, g(x) = x - 3$. Domain of $\underline{f} \cdot g$:

$$\mathbb{R}$$

$$x \neq 3$$

$$x > 3$$

$$x \neq 0$$

$f(x) = \sqrt{x}, g(x) = \sqrt{4 - x}$. Domain of $f \cdot g$:

$$x \geq 0$$

$$x \leq 4$$

$$0 \leq x \leq 4$$

$$\mathbb{R}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sum of functions	Funksiyalar yig'indisi	Сумма функций
Difference of functions	Funksiyalar ayirmasi	Разность функций
Product of functions	Funksiyalar ko'paytmasi	Произведение функций
Quotient of functions	Funksiyalar bo'linmasi	Частное функций
Intersection of sets	To'plamlar kesishmasi	Пересечение множеств
Common domain	Umumiy aniqlanish sohasi	Общая область определения
Zero of a function	Funksiya nuli	Ноль функции
Closed under	Yopiq	Замкнуто относительно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(f \cdot g)(2)$ means:

Dividing functions forbids inputs where:

If both domains are $\mathbb{R}$, the domain of $f - g$ is:

$f(x) = x^2$, $g(x) = -x^2$. Then $f + g$ is:

the zero function

$2x^2$

x^4

undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f = 3x, g = x + 1$; find $(f + g)(2)$

ANSWER 9

2. Same pair; $(f - g)(2)$

ANSWER 3

3. Same pair; $(f \cdot g)(2)$

ANSWER 18

4. Same pair; $(f/g)(2)$

ANSWER 2

5. $f = x^2, g = 4$; find $(f + g)(x)$

ANSWER $x^2 + 4$

6. Same pair; find $(f \cdot g)(x)$

ANSWER $4x^2$

7. Domain of $f + g$ when both are polynomials

ANSWER $\mathbb{R}$

Medium · the standard question

7 PROBLEMS

1. $f = x + 3, g = x - 3$; find $(f \cdot g)(x)$

ANSWER $x^2 - 9$

2. $f = 2x, g = x^2$; solve $(f + g)(x) = 0$

ANSWER $x = 0$ or $x = -2$

3. $f = \sqrt{x}, g = x$; domain of f/g

ANSWER $x > 0$

4. $f = \frac{1}{x}, g = x - 2$; domain of $f + g$

ANSWER $x \neq 0$

5. $f = x - 1, g = x^2 - 1$; simplify f/g

ANSWER $\frac{1}{x+1}, x \neq \pm 1$

6. $f = 5, g = x$; find $(f - g)(-3)$

ANSWER 8

7. $f = x^2, g = x$; solve $(f/g)(x) = 4$

ANSWER $x = 4$

Hard · stretch and reason

7 PROBLEMS

1. $f = \sqrt{x-2}, g = \sqrt{8-x}$; domain of $f+g$

ANSWER $2 \leq x \leq 8$

2. Same pair; domain of f/g

ANSWER $2 \leq x < 8$

3. $f = x^2 - 4, g = x + 2$; simplify f/g with its domain

ANSWER $x - 2, x \neq -2$

4. Find f, g with $(f+g)(x) = x^2$ and $(f-g)(x) = 2x$

ANSWER $f = \frac{x^2 + 2x}{2}, g = \frac{x^2 - 2x}{2}$

5. $f = \frac{1}{x-1}, g = \frac{1}{x+1}$; simplify $f-g$

ANSWER $\frac{2}{x^2 - 1}$

6. Show the sum of two even functions is even

ANSWER $(f+g)(-x) = f(-x)+g(-x) = f(x)+g(x)$

7. Is the product of two odd functions odd or even?

ANSWER even

07 Homework

Homework — 5 tasks

State the domain with every combined function, even when it is $\mathbb{R}$.

1. $f(x) = x + 4, g(x) = x^2 - 16$. Find all four combinations.

2. Give the domain of each of those four.

3. $f(x) = \sqrt{x+1}, g(x) = x - 3$. Find the domain of $\frac{f}{g}$.

4. Solve $(f + g)(x) = 0$ for $f(x) = 2x - 1$, $g(x) = x + 7$.
5. Explain why $\frac{x^2 - 1}{x - 1}$ and $x + 1$ are not the same function.